

7. 小波和多分辨率处理 (Wavelets and Multiresolution Processing)

陆福相

兰州大学信息科学与工程学院

提纲

- 1 背景
- 2 多分辨率展开
- 3 Wavelet Transform in One Dimension
- 4 The Fast Wavelet Transform
- 5 Wavelet Transform in Two Dimensions

- 1 背景
- 2 多分辨率展开
- 3 Wavelet Transform in One Dimension
- 4 The Fast Wavelet Transform
- 5 Wavelet Transform in Two Dimensions

1 背景

- 图像金字塔
- 子带编码
- Haar 变换

2 多分辨率展开

3 Wavelet Transform in One Dimension

4 The Fast Wavelet Transform

5 Wavelet Transform in Two Dimensions

一个图像金字塔是一系列以金字塔形状排列的、分辨率逐渐降低的图像集合。

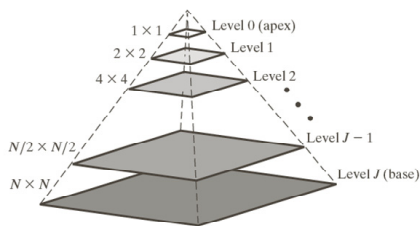
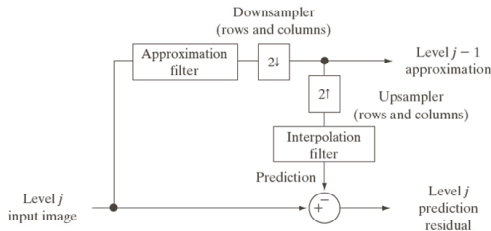


Figure 7.2

(a) 一个图像金字塔；

(b) 创建近似和预测残差金字塔的一个简单系统。



对于 $P > 0$ ，在一个 $P + 1$ 级金字塔中，总像素数是：

$$N^2 \left(1 + \frac{1}{4^1} + \frac{1}{4^2} + \dots + \frac{1}{4^P} \right) \leq \frac{4}{3} N^2$$

近似和预测残差金字塔都以一种迭代方式计算。在第一次迭代之前，以金字塔形式表示的图像放在近似金字塔的第 J 级。然后，下面的 3 步过程执行 P 次， $j = J, J - 1, \dots, J - P + 1$ ：

- **步骤 1** 计算第 j 级输入图像的低分辨率近似。这通过滤波并对滤波结果以 2 为因数下采样完成。将得到的近似放在近似金字塔的第 $j - 1$ 级。
- **步骤 2** 根据步骤 1 得到的低分辨率近似创建第 j 级输入图像的估计（预测）。这通过上采样和滤波完成。
- **步骤 3** 计算步骤 2 得到的预测图像与步骤 1 中输入图像的差，并放在预测残差金字塔的第 j 级。

P 次迭代完成时，第 $J - P$ 级近似输出放到预测残差金字塔的第 $J - P$ 级。

- 近似滤波器
 - 邻域平均，产生均值金字塔；
 - 高斯滤波，产生高斯金字塔；
 - 不滤波，产生子采样金字塔。
- 内插滤波器：nearest neighbor, bilinear, and bicubic.

给定一个整型变量 n 和 1 维样本序列 $f(n)$ ，上采样序列 $f_{2\uparrow}(n)$ 定义如下：

$$f_{2\uparrow}(n) = \begin{cases} f(n/2) & \text{如果 } n \text{ 是偶数} \\ 0 & \text{其他} \end{cases}$$

2 倍下采样定义如下：

$$f_{2\downarrow}(n) = f(2n)$$

例 7.1: 近似和预测残差金字塔。

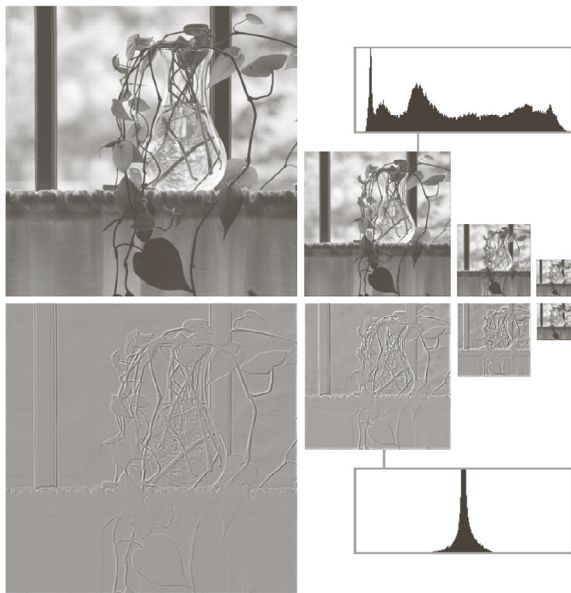


图 7.3 两个图像金字塔和它们的直方图。

(a) 近似金字塔;

(b) 预测残差金字塔; □

1 背景

- 图像金字塔
- 子带编码
- Haar 变换

2 多分辨率展开

3 Wavelet Transform in One Dimension

4 The Fast Wavelet Transform

5 Wavelet Transform in Two Dimensions

一个 FIR (Finite Impulse Response, 有限冲激响应) 滤波器由三个基本部件组成: 延迟单元、乘法器和加法器。

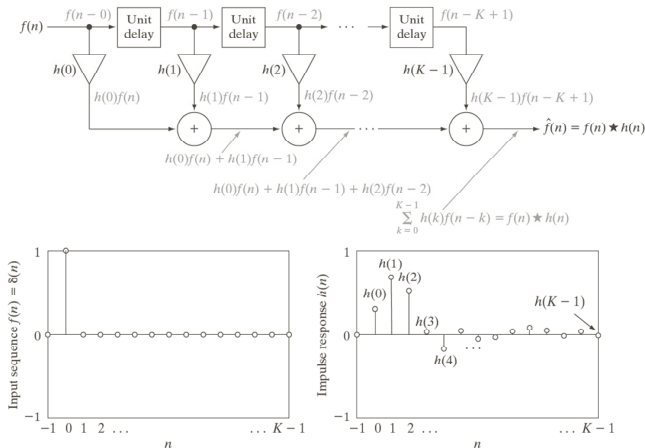


图 7.4

(a) 一个数字滤波器;

(b) 单位离散冲激序列;

(c) 滤波器的冲激响应。

$$\hat{f}(n) = \sum_{k=-\infty}^{\infty} h(k)f(n-k) = f(n) \star h(n)$$

如果 $f(n) = \delta(n)$, 则有

$$\hat{f}(n) = \sum_{k=-\infty}^{\infty} h(k)\delta(n-k) = h(n)$$

- 符号反转: $h_2(n) = -h_1(n)$
- 顺序反转: $h_3(n) = h_1(-n)$ $h_4(n) = h_1(K-1-n)$
- 调制: $h_5(n) = (-1)^n h_1(n)$
- 顺序反转和调制: $h_6(n) = (-1)^n h_1(K-1-n)$

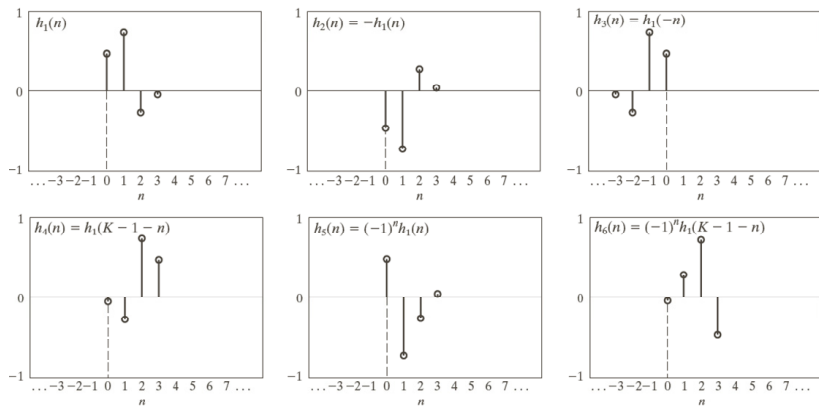


图 7.5 6 个功能上相关的滤波器冲激响应。(a) 参考响应；(b) 符号反转；(c) 和 (d) 顺序反转 (differing by the delay introduced)；(e) 调制；(f) 顺序反转和调制。

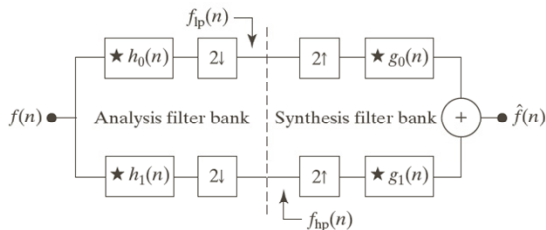
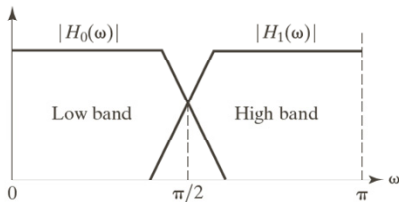


图 7.6

(a) 二子带编码和解码系统:

(b) 它的谱分离 (its spectrum splitting) 特性。



理想重构滤波器组: 选择 $h_0(n)$ 、 $h_1(n)$ 、 $g_0(n)$ 和 $g_1(n)$ 使 $\hat{f}(n) = f(n)$ 。

对于理想重构, $h_0(n)$ 、 $h_1(n)$ 、 $g_0(n)$ 和 $g_1(n)$ 必须满足:

$$g_0(n) = (-1)^n h_1(n)$$

$$g_1(n) = (-1)^{n+1} h_0(n)$$

或

$$g_0(n) = (-1)^{n+1} h_1(n)$$

$$g_1(n) = (-1)^n h_0(n)$$

此外, 它们满足双正交 (**biorthogonality**) 条件:

$$\langle h_i(2n-k), g_j(k) \rangle = \delta(i-j)\delta(n), \quad i, j = \{0, 1\}$$

标准正交 (**Orthonormality**) 滤波器:

$$\langle g_i(n), g_j(n+2m) \rangle = \delta(i-j)\delta(m), \quad i, j = \{0, 1\}$$

标准正交滤波器也满足以下条件:

$$g_1(n) = (-1)^n g_0(K_{\text{even}} - 1 - n)$$

$$h_i(n) = g_i(K_{\text{even}} - 1 - n), \quad i = \{0, 1\}$$

一维标准正交和双正交滤波器可被用于二维可分离滤波器。得到的滤波器输出，记作 $a(m,n)$ 、 $d^V(m,n)$ 、 $d^H(m,n)$ 和 $d^D(m,n)$ ，被称为近似、垂直细节、水平细节和对角线细节子带。

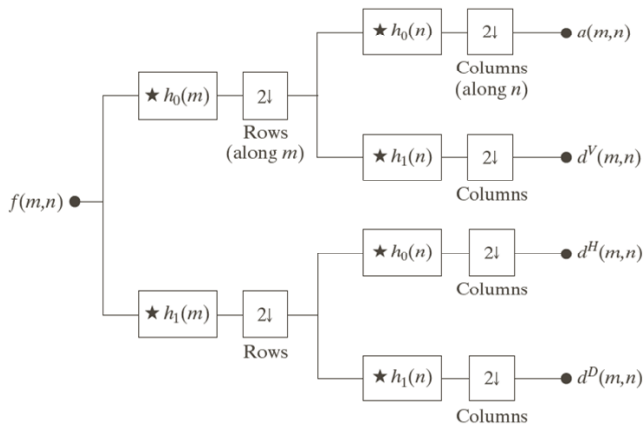


图 7.7 用于子带图像编码的二维、四子带滤波器组。

例 7.2: 花瓶的四子带编码。

n	$g_0(n)$
0	0.23037781
1	0.71484657
2	0.63088076
3	-0.02798376
4	-0.18703481
5	0.03084138
6	0.03288301
7	-0.01059740

表 7.1 $g_0(n)$
的 Daubechies 8-抽头标
准正交滤波器系数。

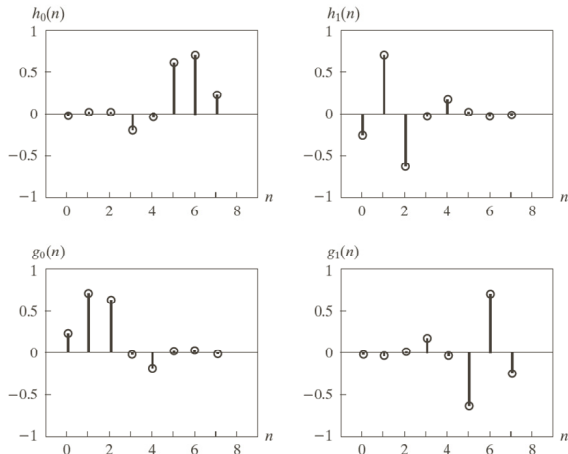


图 7.8 四个 8-抽头 Daubechies 标准正交滤波器的冲激响应。

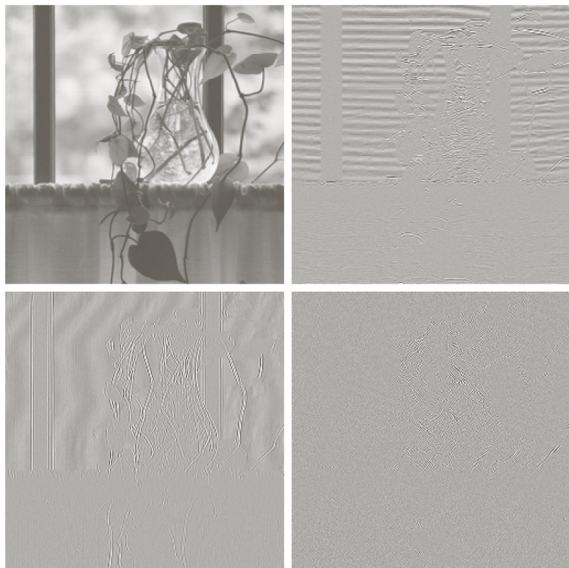


图 7.9 使用图 7.7 所示子带编码系统对图 7.1 的花瓶进行四子带分解。

- (a) 近似子带;
- (b) 水平细节子带;
- (c) 垂直细节子带;
- (d) 对角线细节子带。

1 背景

- 图像金字塔
- 子带编码
- Haar 变换

2 多分辨率展开

3 Wavelet Transform in One Dimension

4 The Fast Wavelet Transform

5 Wavelet Transform in Two Dimensions

Harr 变换的基函数是已知最古老、最简单的标准正交小波。Haar 变换可用下式表示:

$$\mathbf{T} = \mathbf{H}\mathbf{F}\mathbf{H}^T$$

$\mathbf{H}$ 包含 Harr 基函数 $h_k(z)$ 。它们定义在连续闭区间 $z \in [0, 1]$, $k = 0, 1, 2, \dots, N-1$, 其中 $N = 2^n$ 。为了生成 $\mathbf{H}$, 定义整数 k , 即 $k = 2^p + q - 1$, 其中 $0 \leq p \leq n-1$, 当 $p = 0$ 时, $q = 0$ 或 1 , 当 $p \neq 0$ 时, $1 \leq q \leq 2^p$ 。因此, Haar 基函数是

$$h_0(z) = h_{00}(z) = \frac{1}{\sqrt{N}}, \quad z \in [0, 1]$$

和

$$h_k(z) = h_{pq}(z) = \frac{1}{\sqrt{N}} \begin{cases} 2^{p/2} & (q-1)/2^p \leq z < (q-0.5)/2^p \\ -2^{p/2} & (q-0.5)/2^p \leq z < q/2^p \\ 0 & \text{otherwise, } z \in [0, 1] \end{cases}$$

$N \times N$ Haar 变换矩阵的第 i 行包含元素 $h_i(z)$, $z = 0/N, 1/N, 2/N, \dots, (N-1)/N$ 。

$$\mathbf{H}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad \mathbf{H}_4 = \frac{1}{\sqrt{4}} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ \sqrt{2} & -\sqrt{2} & 0 & 0 \\ 0 & 0 & \sqrt{2} & \sqrt{2} \end{bmatrix}$$

例 7.3: 离散小波变换中的 Harr 函数。

- 除了图 7.10(a) 左上角子图像，局部直方图非常相似。许多像素值接近于零。分解中的大量零值使得图像对于压缩来说是一个优秀的候选者。
- 图 7.10(a) 的子图像可用来重构原始花瓶图像的粗、细分辨率近似。
- 以迭代方式分解。
- 图 7.10(a) 不是花瓶图像的 Haar 变换。
- 每个子图像表示原始图像中空间频率的一个特定频带。

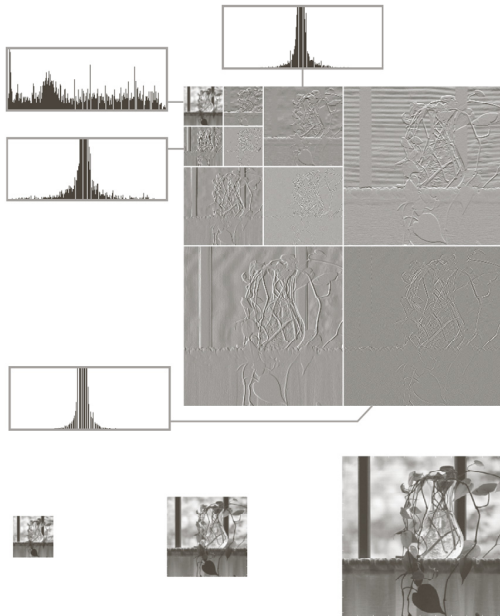


Figure 7.10

(a) 使用 Haar H_2 基函数的离散小波变换。图中也显示了局部直方图变化。

(b)-(d) 由 (a) 得到的几种不同的近似 (64×64 、 128×128 和 256×256)。 □

- 1 背景
- 2 多分辨率展开
- 3 Wavelet Transform in One Dimension
- 4 The Fast Wavelet Transform
- 5 Wavelet Transform in Two Dimensions

- 1 背景
- 2 多分辨率展开
 - 级数展开
 - Scaling Functions
 - Wavelet Functions
- 3 Wavelet Transform in One Dimension
- 4 The Fast Wavelet Transform
- 5 Wavelet Transform in Two Dimensions

$$f(x) = \sum_k \alpha_k \varphi_k(x)$$

where the α_k are called *expansion coefficients*, the $\varphi_k(x)$ are called *expansion functions*. If the expansion is unique – that is, there is only one set of α_k for any given $f(x)$ – the $\varphi_k(x)$ are called *basis functions*, and the expansion set, $\{\varphi_k(x)\}$, is called a *basis* for the class of functions that can be so expressed.

The *closed span* (i.e., the function space) of the expansion set is denoted as

$$V = \overline{\text{Span}_k \{\varphi_k(x)\}}$$

$$\alpha_k = \langle \tilde{\varphi}_k(x), f(x) \rangle = \int \tilde{\varphi}_k^*(x) f(x) dx$$

- Case 1 (the orthonormal basis). If

$$\langle \varphi_j(x), \varphi_k(x) \rangle = \delta_{jk} = \begin{cases} 0 & j \neq k \\ 1 & j = k \end{cases}$$

then $\varphi_k(x) = \tilde{\varphi}_k(x)$. So, $\alpha_k = \langle \varphi_k(x), f(x) \rangle$.

- Case 2 (the orthogonal basis). If

$$\langle \varphi_j(x), \varphi_k(x) \rangle = 0 \quad j \neq k$$

then the basis functions and their dual are called *biorthogonal*

$$\langle \varphi_j(x), \tilde{\varphi}_k(x) \rangle = \delta_{jk} = \begin{cases} 0 & j \neq k \\ 1 & j = k \end{cases}$$

- **Case 3 (not a basis).** If $\{\varphi_k(x)\}$ is a spanning set in which there is more than one set of α_k for any $f(x) \in V$. The expansion functions and their duals are said to be *overcomplete* or *redundant*. They form a *frame* in which

$$A\|f(x)\|^2 \leq \sum_k |\langle \varphi_k(x), f(x) \rangle|^2 \leq B\|f(x)\|^2$$

for some $A > 0$, $B < \infty$, and all $f(x) \in V$. If $A = B$, the expansion set is called a *tight frame* and it can be shown that

$$f(x) = \frac{1}{A} \sum_k \langle \varphi_k(x), f(x) \rangle \varphi_k(x)$$

- 1 背景
- 2 多分辨率展开
 - 级数展开
 - **Scaling Functions**
 - Wavelet Functions
- 3 Wavelet Transform in One Dimension
- 4 The Fast Wavelet Transform
- 5 Wavelet Transform in Two Dimensions

Consider the set $\{\varphi_{j,k}(x)\}$, where

$$\varphi_{j,k}(x) = 2^{j/2} \varphi(2^j x - k)$$

for all $j, k \in \mathbb{Z}$ and $\varphi(x) \in L^2(\mathbf{R})$. Because the shape of $\varphi_{j,k}(x)$ changes with j , $\varphi(x)$ is called a *scaling function*. By choosing $\varphi(x)$ properly, $\{\varphi_{j,k}(x)\}$ can be made to span $L^2(\mathbf{R})$, which is the set of all measurable, square-integrable functions.

We denote the subspace spanned over k for any j as

$$V_j = \overline{\text{Span}_k \{\varphi_{j,k}(x)\}}$$

Increasing j increases the size of V_j , allowing functions with smaller variations or finer detail to be included in the subspace.

Example 7.4 The Haar scaling function.

$$\varphi(x) = \begin{cases} 1 & 0 \leq x < 1 \\ 0 & \text{otherwise} \end{cases}$$

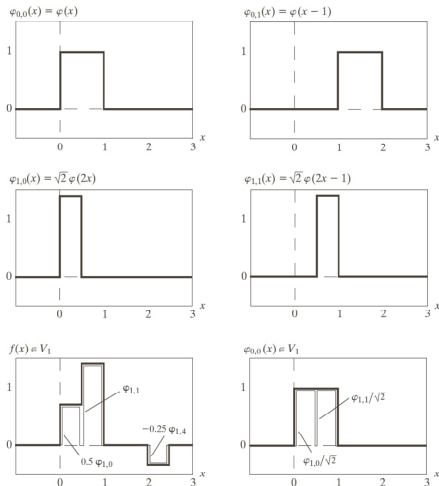


Figure 7.11 Some Harr scaling functions. □

$$f(x) = 0.5\varphi_{1,0}(x) + \varphi_{1,1}(x) - 0.25\varphi_{1,4}(x)$$

$$\varphi_{0,k}(x) = \frac{1}{\sqrt{x}}\varphi_{1,2k}(x) + \frac{1}{\sqrt{2}}\varphi_{1,2k+1}(x)$$

$$V_0 \subset V_1$$

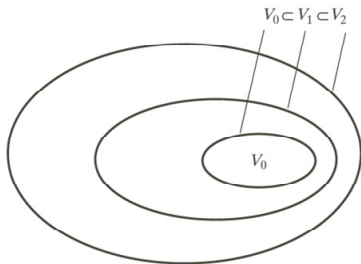


Figure 7.12 The nested function spaces spanned by a scaling function.

The four fundamental requirements of multiresolution analysis:

- ① The scaling function is orthogonal to its integer translates. The Haar scaling function is said to have compact support, which means that it is 0 everywhere outside a finite interval called the *support*.
- ② The subspaces spanned by the scaling function at low scaling are nested within those spanned at higher scales (see Fig. 7.12).

$$V_{-\infty} \subset \cdots \subset V_{-1} \subset V_0 \subset V_1 \subset \cdots \subset V_{\infty}$$

Moreover, the subspaces satisfy the intuitive condition that if $f(x) \in V_j$, then $f(2x) \in V_{j+1}$. Note that

$$\varphi(x) = \begin{cases} 1 & 0.25 \leq x \leq 0.75 \\ 0 & \text{elsewhere} \end{cases}$$

is not a valid scaling function.

- ③ The only function that is common to all V_j is $f(x) = 0$.
 $V_{-\infty} = \{0\}$.
- ④ Any function can be represented with arbitrary precision.
 $V_{\infty} = \{L^2(\mathbf{R})\}$.

$$\begin{aligned}
\varphi_{j,k}(x) &= \sum_n \alpha_n \varphi_{j+1,n}(x) \\
\Rightarrow \varphi_{j,k}(x) &= \sum_n h_\varphi(n) 2^{(j+1)/2} \varphi(2^{j+1}x - n) \\
\Rightarrow \varphi(x) &\stackrel{*}{=} \sum_n h_\varphi(n) \sqrt{2} \varphi(2x - n)
\end{aligned}$$

$h_\varphi(n)$: scaling function coefficients, h_φ : scaling vector. Equation (*) is called the *refinement equation*, the *MRA equation*, or the *dilation equation*.

Example 7.5 Haar scaling function coefficients.

$$\varphi(x) = \frac{1}{\sqrt{2}} \left[\sqrt{2} \varphi(2x) \right] + \frac{1}{\sqrt{2}} \left[\sqrt{2} \varphi(2x - 1) \right]$$

$h_\varphi(0) = h_\varphi(1) = 1/\sqrt{2}$, which are the first row of matrix $\mathbf{H}_2$. □

- 1 背景
- 2 多分辨率展开
 - 级数展开
 - Scaling Functions
 - **Wavelet Functions**
- 3 Wavelet Transform in One Dimension
- 4 The Fast Wavelet Transform
- 5 Wavelet Transform in Two Dimensions

Given a scaling function that meets the MRA requirements, we can define a *wavelet function* $\psi(x)$ that, together with its integer translates and binary scaling, spans the difference between any two adjacent scaling subspaces, V_j and V_{j+1} .

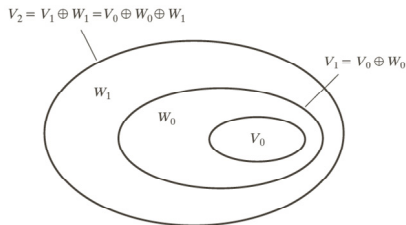


Figure 7.13 The relationship between scaling and wavelet function spaces.

$$\psi_{j,k}(x) = 2^{j/2} \psi(2^j x - k)$$

$$W_j = \overline{\text{Span}_k \{ \psi_{j,k}(x) \}}$$

$$f(x) = \sum_k \alpha_k \psi_{j,k}(x), \text{ if } f(x) \in W_j$$

$$V_{j+1} = V_j \oplus W_j$$

The orthogonal complement of V_j in V_{j+1} is W_j , and all members of V_j are orthogonal to the members of W_j . Thus

$$\langle \varphi_{j,k}(x), \psi_{j,l}(x) \rangle = 0$$

for all appropriate $j, k, l \in \mathbf{Z}$.

We can express the space of all measurable, square-integrable functions as

$$L^2(\mathbf{R}) = \cdots \oplus W_{-2} \oplus W_{-1} \oplus W_0 \oplus W_1 \oplus W_2 \oplus \cdots$$

$$L^2(\mathbf{R}) = V_{j_0} \oplus W_{j_0} \oplus W_{j_0+1} \oplus W_{j_0+2} \oplus \cdots$$

$$\psi(x) = \sum_n h_\psi(n) \sqrt{2} \varphi(2x - n)$$

where the $h_\psi(n)$ are called the *wavelet function coefficients* and h_ψ is the *wavelet vector*. We can also show that $h_\psi(n)$ is related to $h_\varphi(n)$ by

$$h_\psi(n) = (-1)^n h_\varphi(1 - n)$$

Example 7.6: The Haar wavelet function coefficients.

$h_\varphi(0) = h_\varphi(1) = 1/\sqrt{2}$, $h_\psi(0) = 1/\sqrt{2}$, and $h_\psi(1) = -1/\sqrt{2}$. Note that these coefficients correspond to the second row of matrix $\mathbf{H}_2$. We can get $\psi(x) = \varphi(2x) - \varphi(2x - 1)$.

$$\psi(x) = \begin{cases} 1 & 0 \leq x < 0.5 \\ -1 & 0.5 \leq x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

$$f(x) = f_a(x) + f_d(x)$$

where

$$f_a(x) = \frac{3\sqrt{2}}{4}\varphi_{0,0}(x) - \frac{\sqrt{2}}{8}\varphi_{0,2}(x)$$

and

$$f_d(x) = \frac{-\sqrt{2}}{4}\psi_{0,0}(x) - \frac{\sqrt{2}}{8}\psi_{0,2}(x)$$

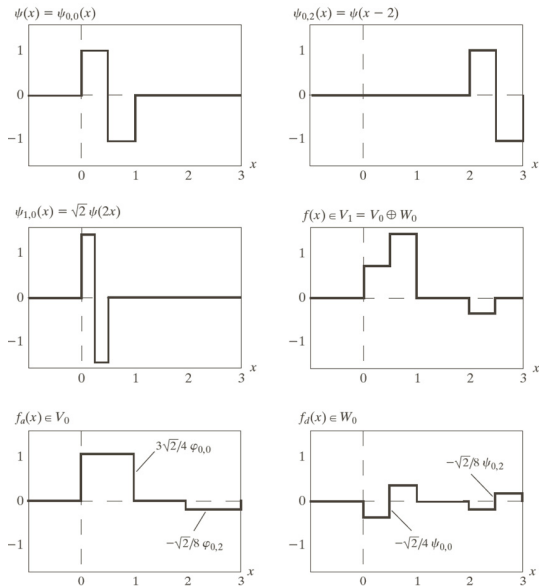


Figure 7.14 Harr wavelet functions in W_0 and W_1 .



- 1 背景
- 2 多分辨率展开
- 3 Wavelet Transform in One Dimension
- 4 The Fast Wavelet Transform
- 5 Wavelet Transform in Two Dimensions

- 1 背景
- 2 多分辨率展开
- 3 Wavelet Transform in One Dimension
 - The Wavelet Series Expansion
 - The Discrete Wavelet Transform
 - The Continuous Wavelet Transform
- 4 The Fast Wavelet Transform
- 5 Wavelet Transform in Two Dimensions

$$f(x) = \sum_k c_{j_0}(k) \varphi_{j_0,k}(x) + \sum_{j=j_0}^{\infty} \sum_k d_j(k) \psi_{j,k}(x)$$

where j_0 is an arbitrary starting scale. The $c_{j_0}(k)$ are normally called *approximation* and/or *scaling coefficients*; the $d_j(k)$ are referred to as *detail* and/or *wavelet coefficients*. If the expansion functions form an orthonormal basis or tight frame, the expansion coefficients are calculated as

$$c_{j_0}(k) = \langle f(x), \varphi_{j_0,k}(x) \rangle = \int f(x) \varphi_{j_0,k}(x) dx$$

$$d_j(k) = \langle f(x), \psi_{j,k}(x) \rangle = \int f(x) \psi_{j,k}(x) dx$$

If the expansion functions are part of a biorthogonal basis, the φ and ψ terms in the preceding two equations must be replaced by their dual functions, $\tilde{\varphi}$ and $\tilde{\psi}$, respectively.

Example 7.7: The Haar wavelet series expansion of $y = x^2$.

$$y = \begin{cases} x^2 & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Using Haar wavelets and $j_0 = 0$, the expansion coefficients are computed as:

$$c_0(0) = \int_0^1 x^2 \varphi_{0,0} dx = \int_0^1 x^2 dx = \frac{1}{3}$$

$$d_0(0) = \int_0^1 x^2 \psi_{0,0} dx = -\frac{1}{4}$$

$$d_1(0) = \int_0^1 x^2 \psi_{1,0} dx = -\frac{\sqrt{2}}{32}$$

$$d_1(1) = \int_0^1 x^2 \psi_{1,1} dx = -\frac{3\sqrt{2}}{32}$$

$$\begin{aligned}
 y = & \underbrace{\frac{1}{3}\varphi_{0,0}(x)}_{V_0} + \underbrace{\left[-\frac{1}{4}\psi_{0,0}(x)\right]}_{W_0} + \left[-\frac{\sqrt{2}}{32}\psi_{1,0}(x) - \frac{3\sqrt{2}}{32}\psi_{1,1}(x)\right] + \cdots \\
 & \underbrace{\hspace{10em}}_{V_1=V_0\oplus W_0} \\
 & \underbrace{\hspace{15em}}_{V_2=V_1\oplus W_1=V_0\oplus W_0\oplus W_1}
 \end{aligned}$$

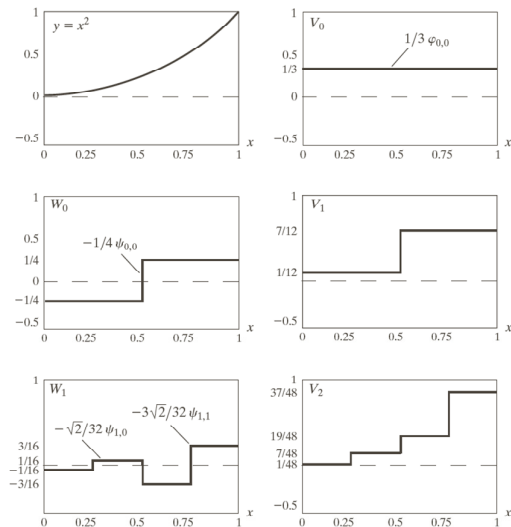


Figure 7.15 A wavelet series expansion of $y = x^2$ using Haar wavelets. □

- 1 背景
- 2 多分辨率展开
- 3 Wavelet Transform in One Dimension
 - The Wavelet Series Expansion
 - The Discrete Wavelet Transform
 - The Continuous Wavelet Transform
- 4 The Fast Wavelet Transform
- 5 Wavelet Transform in Two Dimensions

The forward DWT coefficients for sequence $f(n)$ are

$$W_{\varphi}(j_0, k) = \frac{1}{\sqrt{M}} \sum_n f(n) \varphi_{j_0, k}(n)$$
$$W_{\psi}(j, k) = \frac{1}{\sqrt{M}} \sum_n f(n) \psi_{j, k}(n) \quad \text{for } j \geq j_0$$

The inverse DWT is

$$f(n) = \frac{1}{\sqrt{M}} \sum_k W_{\varphi}(j_0, k) \varphi_{j_0, k}(n) + \frac{1}{\sqrt{M}} \sum_{j=j_0}^{\infty} \sum_k W_{\psi}(j, k) \psi_{j, k}(n)$$

Normally, we let $j_0 = 0$ and select M to be a power of 2 (i.e., $M = 2^J$) so that the summations are performed over $n = 0, 1, 2, \dots, M - 1$, $j = 0, 1, 2, \dots, J - 1$, and $k = 0, 1, 2, \dots, 2^j - 1$. The transform itself is composed of M coefficients, the minimum scale is 0, and the maximum scale is $J - 1$.

For Haar wavelets, the discretized scaling and wavelet functions employed in the transform (i.e, the basis functions) correspond to the rows of the $M \times M$ of Haar transformation matrix.

Example 7.8: Computing a one-dimensional discrete wavelet transform.

$f(0) = 1$, $f(1) = 4$, $f(2) = -3$, and $f(3) = 0$. $M = 4$, $J = 2$.

$x = 0, 1, 2, 3$. $k = 0$ for $j_0 = 0$. $k = 0$ for $j = 0$ or $k = 0, 1$ for $j = 1$.

$$W_{\varphi}(0,0) = \frac{1}{2} \sum_{n=0}^3 f(n) \varphi_{0,0}(n) = \frac{1}{2} [1 \cdot 1 + 4 \cdot 1 - 3 \cdot 1 + 0 \cdot 1] = 1$$

$$W_{\psi}(0,0) = \frac{1}{2} [1 \cdot 1 + 4 \cdot 1 - 3 \cdot (-1) + 0 \cdot (-1)] = 4$$

$$W_{\psi}(1,0) = \frac{1}{2} [1 \cdot \sqrt{2} + 4 \cdot (-\sqrt{2}) - 3 \cdot 0 + 0 \cdot 0] = -1.5\sqrt{2}$$

$$W_{\psi}(1,1) = \frac{1}{2} [1 \cdot 0 + 4 \cdot 0 - 3 \cdot \sqrt{2} + 0 \cdot (-\sqrt{2})] = -1.5\sqrt{2}$$

$$f(n) = \frac{1}{2} W_{\varphi}(0,0) \varphi_{0,0}(n) + W_{\psi}(0,0) \psi_{0,0}(n) + W_{\psi}(1,0) \psi_{1,0}(n) \\ + W_{\psi}(1,1) \psi_{1,1}(n) \quad \text{for } n = 0, 1, 2, 3$$

$$f(0) = \frac{1}{2} [1 \cdot 1 + 4 \cdot 1 - 1.5\sqrt{2} \cdot \sqrt{2} - 1.5\sqrt{2} \cdot 0] = 1 \quad \square$$

- 1 背景
- 2 多分辨率展开
- 3 Wavelet Transform in One Dimension
 - The Wavelet Series Expansion
 - The Discrete Wavelet Transform
 - The Continuous Wavelet Transform
- 4 The Fast Wavelet Transform
- 5 Wavelet Transform in Two Dimensions

The CWT is defined as

$$W_{\psi}(s, \tau) = \int_{-\infty}^{\infty} f(x) \psi_{s, \tau}(x) dx$$

where

$$\psi_{s, \tau}(x) = \frac{1}{\sqrt{s}} \psi\left(\frac{x - \tau}{s}\right)$$

and s and τ are called *scale* and *translation* parameters.

Given $W_{\psi}(s, \tau)$, $f(x)$ can be obtained using the *inverse* CWT

$$f(x) = \frac{1}{C_{\psi}} \int_0^{\infty} \int_{-\infty}^{\infty} W_{\psi}(s, \tau) \frac{\psi_{s, \tau}(x)}{s^2} d\tau ds$$

where

$$C_{\psi} = \int_{-\infty}^{\infty} \frac{|\Psi(\mu)|^2}{|\mu|} d\mu$$

and $\Psi(\mu) = \mathfrak{F}\{\psi(x)\}$. The four preceding equations define a reversible transform as long as the so-called *admissibility criterion*, $C_{\psi} < \infty$, is satisfied. In most cases, this simply means that $\Psi(0) = 0$ and $\Psi(\mu) \rightarrow 0$ as $\mu \rightarrow \infty$ fast enough to make $C_{\psi} < \infty$.

Example 7.9: A one-dimensional continuous wavelet transform.

The *Mexican hat* wavelet

$$\psi(x) = \left(\frac{2}{\sqrt{3}} \pi^{-1/4} (1 - x^2) e^{-x^2/2} \right)$$

The continue, one-dimensional function in Fig. 7.16(a) is the sum of two Mexican hat wavelets:

$$f(x) = \psi_{1,10}(x) + \psi_{6,80}(x)$$

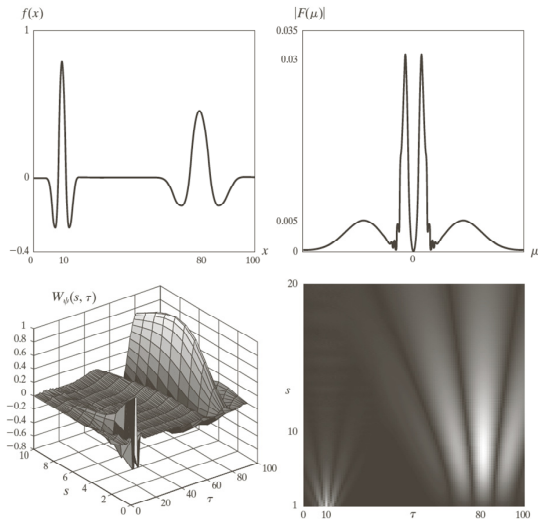


Figure 7.16 The continuous wavelet transform (c and d) and Fourier spectrum (b) of a continuous 1-D function (a). □

- 1 背景
- 2 多分辨率展开
- 3 Wavelet Transform in One Dimension
- 4 The Fast Wavelet Transform**
- 5 Wavelet Transform in Two Dimensions

Consider the multiresolution refinement equation

$$\varphi(x) = \sum_n h_\varphi(n) \sqrt{2} \varphi(2x - n)$$

Scaling x by 2^j , translating it by k , and letting $m = 2k + n$ gives

$$\varphi(2^j x - k) = \sum_n h_\varphi(m - 2k) \sqrt{2} \varphi(2^{j+1} x - m)$$

Similarly, we can obtain

$$\psi(2^j x - k) = \sum_n h_\psi(m - 2k) \sqrt{2} \varphi(2^{j+1} x - m)$$

From the wavelet series expansion, we get

$$\begin{aligned} d_j(k) &= \int f(x) 2^{j/2} \psi(2^j x - k) dx \\ &= \int f(x) 2^{j/2} \left[\sum_m h_\psi(m - 2k) \sqrt{2} \varphi(2^{j+1} x - m) \right] dx \\ &= \sum_m h_\psi(m - 2k) c_{j+1}(m) \end{aligned}$$

Similarly, we find that

$$c_j(k) = \sum_m h_\varphi(m - 2k) c_{j+1}(m)$$

When $f(x)$ is discrete, we can write

$$W_\psi(j, k) = \sum_m h_\psi(m - 2k) W_\varphi(j + 1, m)$$

$$W_\varphi(j, k) = \sum_m h_\varphi(m - 2k) W_\varphi(j + 1, m)$$

or

$$\begin{aligned} W_\psi(j, k) &= h_\psi(-n) \star W_\varphi(j + 1, n)|_{n=2k, k \geq 0} \\ W_\varphi(j, k) &= h_\varphi(-n) \star W_\varphi(j + 1, n)|_{n=2k, k \geq 0} \end{aligned} \quad (1)$$

For a sequence of length $M = 2^J$, the number of mathematical operations involved is on the order of $O(M)$. The FFT algorithm requires on the order of $O(M \log_2 M)$ operations.

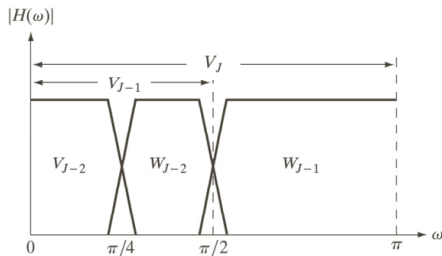
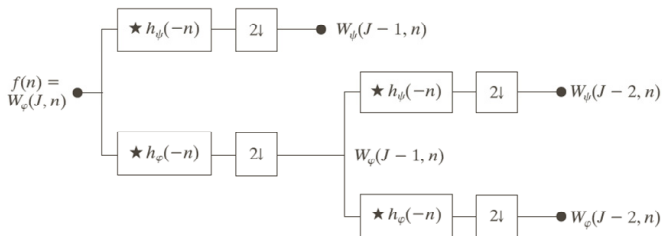


Figure 7.18 (a) A two-stage or two-scale FWT analysis bank and (b) its frequency splitting characteristics.

Example 7.10: Computing a 1-D fast wavelet transform.

$f(n) = \{1, 4, -3, 0\}$. We compute the two-scale FWT based on Haar scaling and wavelet vectors:

$$h_{\varphi}(n) = \begin{cases} 1/\sqrt{2} & n = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

and

$$h_{\psi}(n) = \begin{cases} 1/\sqrt{2} & n = 0 \\ -1/\sqrt{2} & n = 1 \\ 0 & \text{otherwise} \end{cases}$$

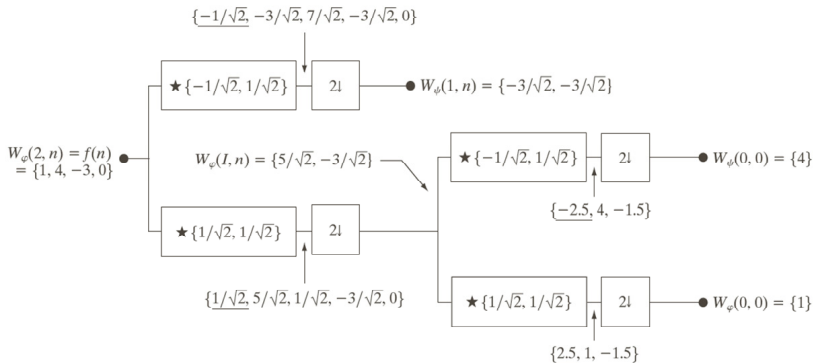


Figure 7.19 Computing a two-scale FWT of sequence $\{1, 4, -3, 0\}$ using Haar scaling and wavelet vectors.

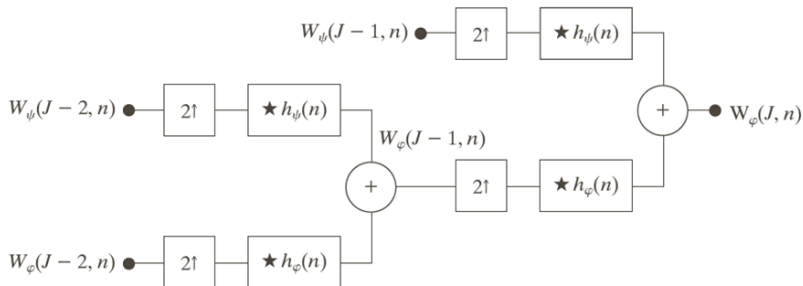


Figure 7.21 A two-stage or two-scale FWT^{-1} synthesis bank.

$$W_{\varphi}(j+1, k) = h_{\varphi}(k) \star W_{\varphi}^{2\uparrow}(j, k) + h_{\psi}(k) \star W_{\psi}^{2\uparrow}(j, k) |_{k \geq 0}$$

Example 7.11: Computing a 1-D inverse FWT.

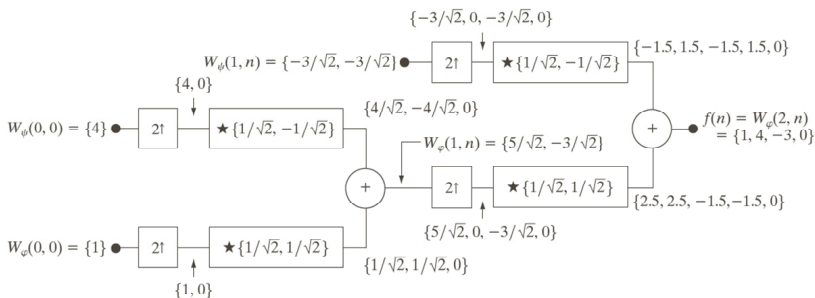


Figure 7.22 Computing a two-scale FWT of sequence $\{1, 4, -3, 0\}$ using Haar scaling and wavelet vectors.

Heisenberg uncertainty principle: if you want precise information about time, you must accept some vagueness about frequency, and vice versa.

To illustrate the principle graphically, each basis function used in the representation of a function can be viewed schematically as a *tile* in a *tile-frequency plane*. The tile, also called a *Heisenberg cell* or *Heisenberg box*, shows the frequency content of the basis function that it represents and where the basis function resides in time.

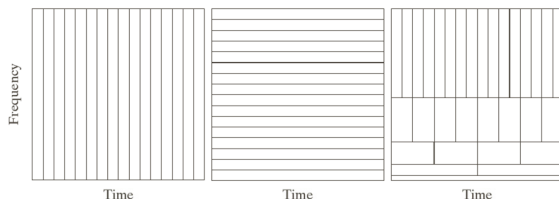


Figure 7.23 Time-frequency tilings for the basis functions associated with (a) sampled data. (b) the FFT, and (c) the FWT. Note that the horizontal strips of equal height rectangle in (c) represent FWT scales.

- 1 背景
- 2 多分辨率展开
- 3 Wavelet Transform in One Dimension
- 4 The Fast Wavelet Transform
- 5 Wavelet Transform in Two Dimensions

The separable scaling and “directionally sensitive” functions are

$$\begin{aligned}\varphi(x, y) &= \varphi(x)\varphi(y) \\ \psi^H(x, y) &= \psi(x)\varphi(y) \\ \psi^V(x, y) &= \varphi(x)\psi(y) \\ \psi^D(x, y) &= \psi(x)\psi(y)\end{aligned}$$

The scaled and translated basis functions are

$$\begin{aligned}\varphi_{j,m,n}(x, y) &= 2^j \varphi(2^j x - m, 2^j y - n) \\ \psi_{j,m,n}^i(x, y) &= 2^j \psi^i(2^j x - m, 2^j y - n), \quad i = \{H, V, D\}\end{aligned}$$

The DWT of image $f(x, y)$ of size $M \times N$ is

$$\begin{aligned}W_\varphi(j_0, m, n) &= \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \varphi_{j_0, m, n}(x, y) \\ W_\psi^i(j, m, n) &= \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \psi_{j, m, n}^i(x, y), \quad i = \{H, V, D\}\end{aligned}$$

We normally let $j_0 = 0$ and select $N = M = 2^J$ so that $j = 0, 1, 2, \dots, J - 1$ and $m = n = 0, 1, 2, \dots, 2^j - 1$.

The inverse DWT is

$$f(x, y) = \frac{1}{\sqrt{MN}} \sum_{m=0} \sum_{n=0} W_{\varphi}(j_0, m, n) \varphi_{j_0, m, n}(x, y) \\ + \frac{1}{\sqrt{MN}} \sum_{i=H, V, D} \sum_{j=j_0}^{\infty} \sum_{m=0} \sum_{n=0} W_{\psi}^i(j, m, n) \psi_{j, m, n}^i(x, y)$$

With separable two-dimensional scaling and wavelet functions, we simply take the 1-D FWT of the rows of $f(x, y)$, followed by the 1-D FWT of the resulting columns.

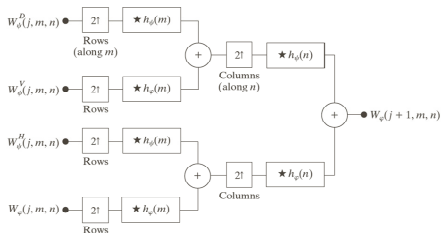
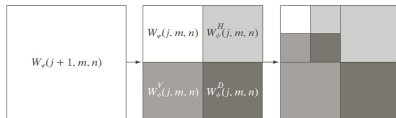
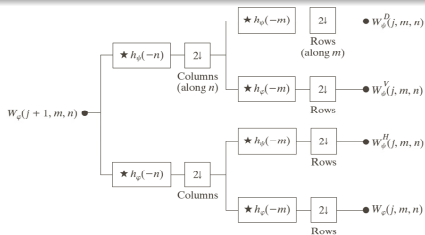


Figure 7.24 The 2-D fast wavelet transform: (a) the analysis filter band; (b) the resulting decomposition; (c) the synthesis filter bank.

Example 7.12: Computing a 2-D fast wavelet transform.

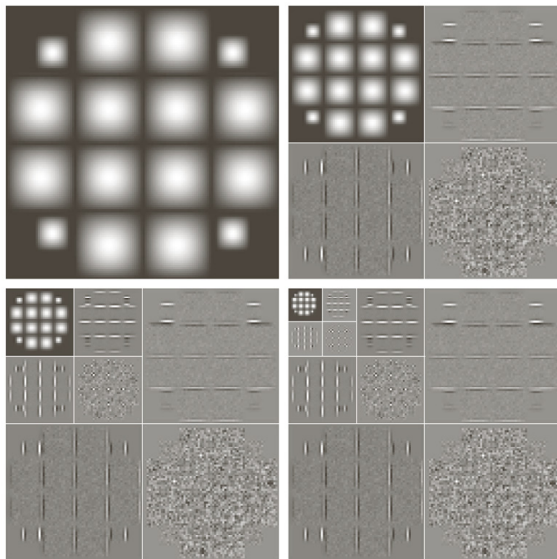


Figure 7.25 Computing a 2-D three-scale FWT: (a) the original image; (b) a one-scale FWT; (c) a two-scale FWT; and (d) a three-scale FWT. $\square$

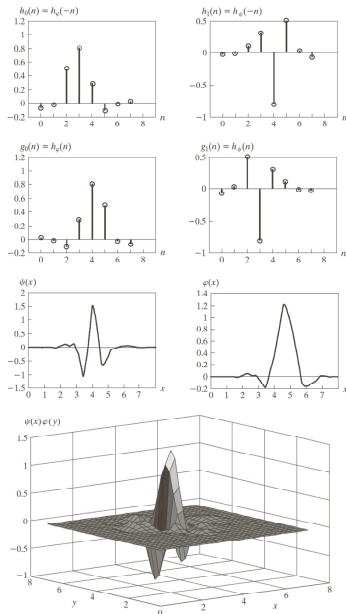


Figure 7.26 Fourth-order symlets: (a)-(b) decomposition filters; (c)-(d) reconstruction filters; (e) the one-dimensional wavelet; (f) the one-dimensional scaling function and (g) one of two-dimensional wavelet, $\psi^V(x, y)$. See Table 7.3 for the values of $h_\varphi(n)$ for $0 \leq n \leq 7$.

n	$h_{\varphi}(n)$
0	0.0322
1	-0.0126
2	-0.0992
3	0.2979
4	0.8037
5	0.4976
6	-0.0296
7	-0.0758

Table 7.3

Orthonormal
fourth-order
symlet filter
coefficients for
 $h_{\varphi}(n)$.

The basic approach of applying wavelet transform to image processing is to

- Compute a 2-D WT of an image;
- Alter the transform;
- Compute the inverse transform.

Example 7.13: Wavelet-based edge detection.

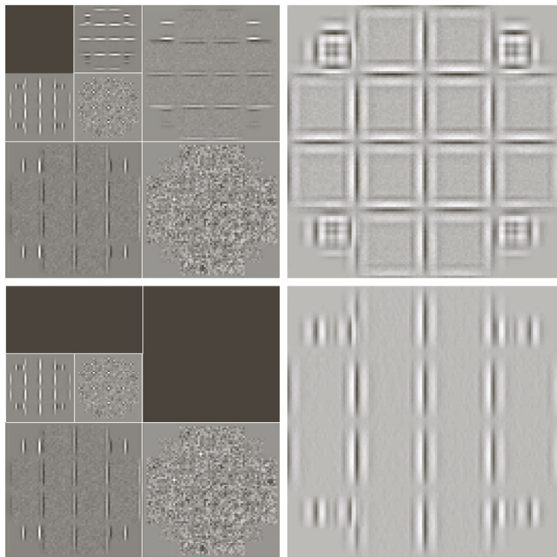


Figure 7.27 Modifying a DWT for edge detection: (a) and (c) two-scale decompositions with selected coefficients deleted; (b) and (d) the corresponding reconstructions.

Example 7.14: Wavelet-based noise removed.

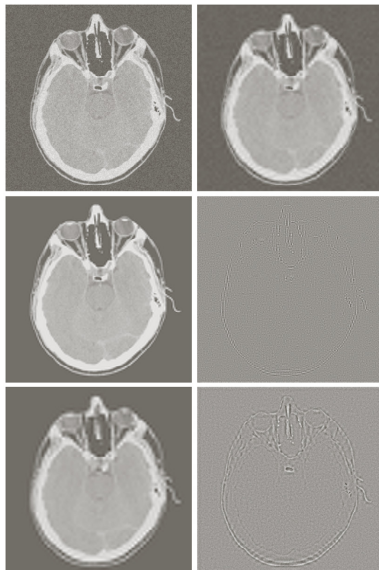


Figure 7.28 Modifying a DWT for noise removal: (a) a noise CT of a human head; (b), (c) and (e) various reconstructions after thresholding the detail coefficients; (d) and (f) the information removed during the reconstruction of (c) and (e). □

Homework

Ex 7.2, 7.8, 7.16, 7.23.